Build Your Own OG Collector

The Birth of AnythingHound

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NorthSec

Hands On

Lab 1
Lab 2
Open Hours

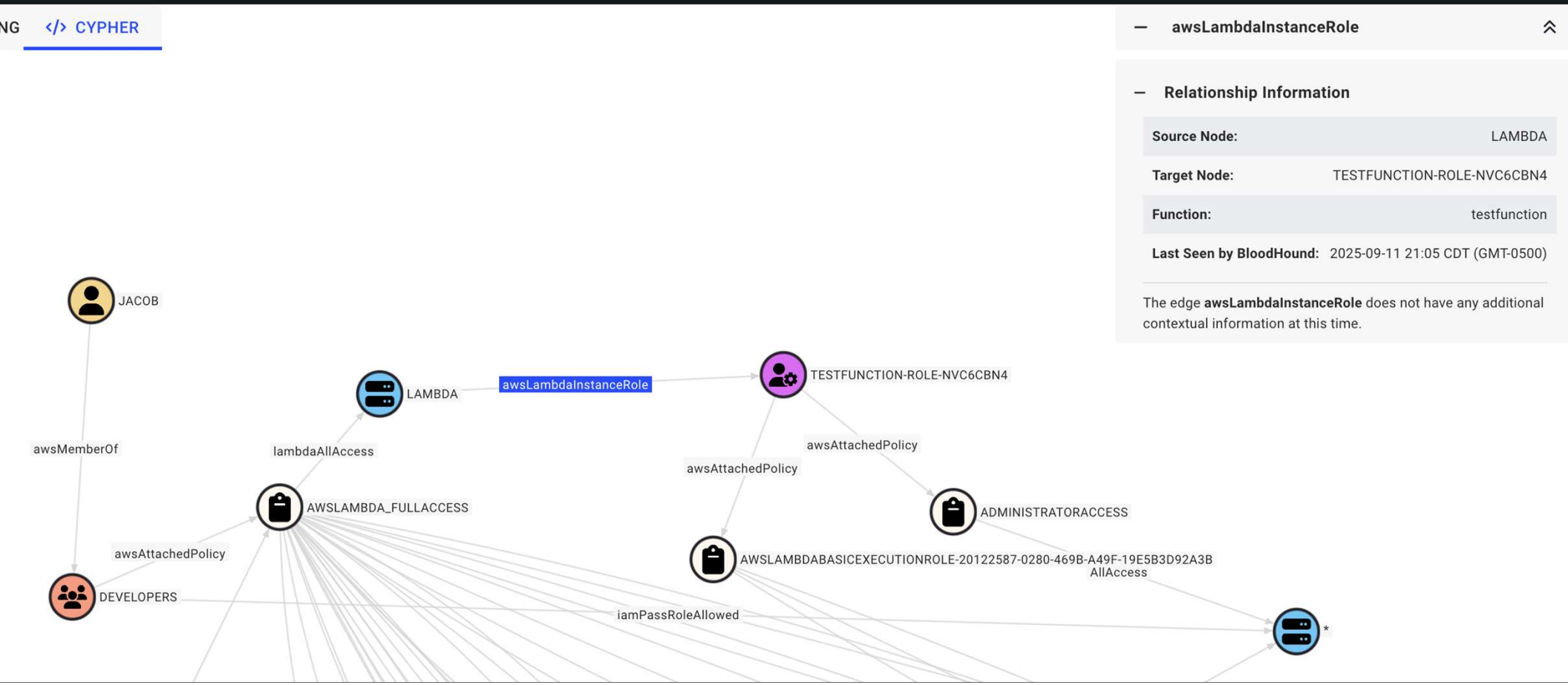
https://github.com/Mayyhem/OpenGraphWorkshop

OpenGraph

What is OpenGraph?

A way to ingest custom files to visualize Attack Paths for any technology

What does it look like?



How Does It Work?

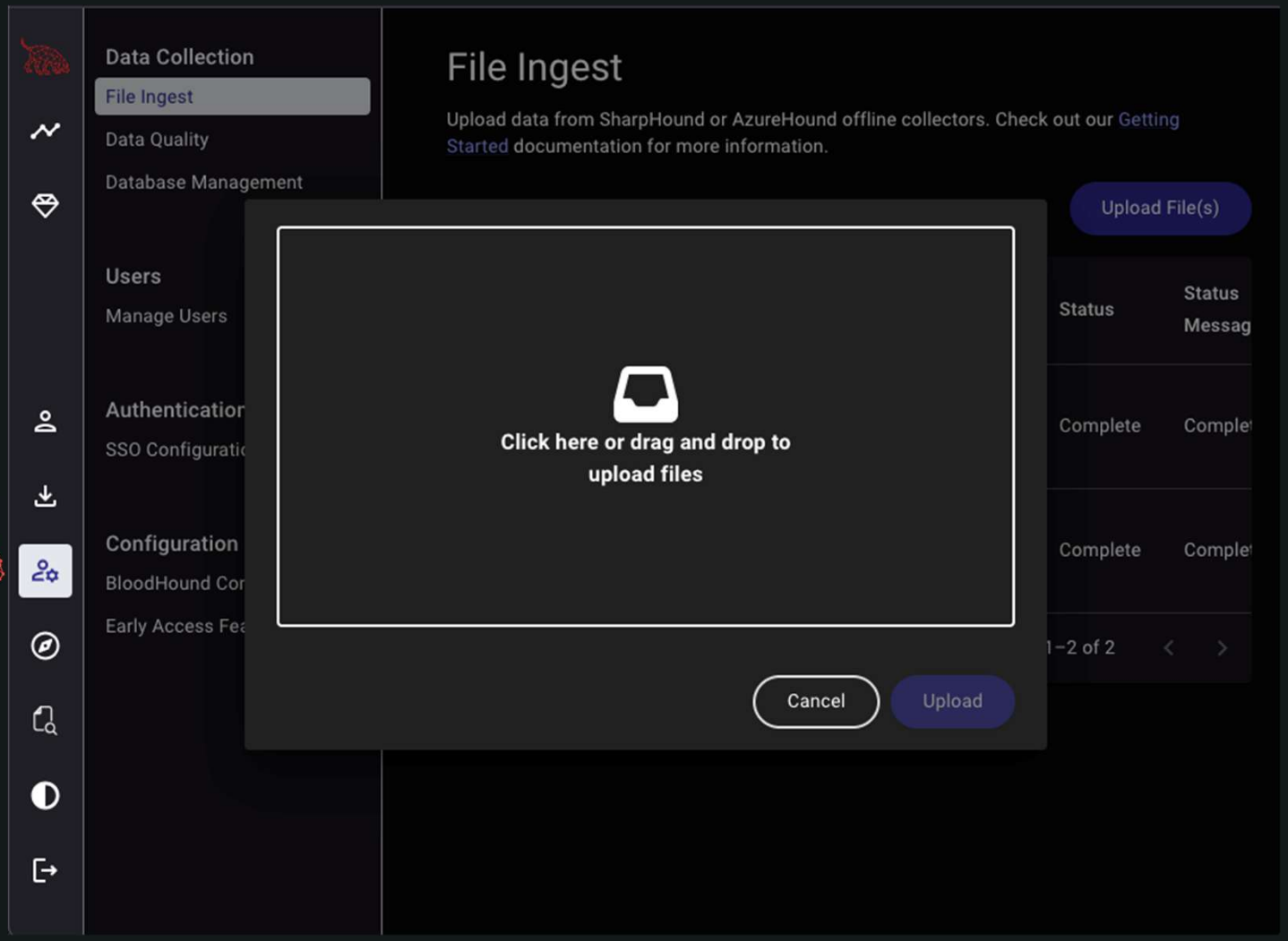
Create a JSON

```json
{
  "graph": {
    "nodes": [
      {
        "id": "123",
        "kinds": [
          "Person",
          "Base"
        ],
        "properties": {
          "displayname": "bob",
          "property": "a",
          "name": "BOB"
        }
      },
      {
        "id": "234",
        "kinds": [
          "Person",
          "Base"
        ],
        "properties": {
          "displayname": "alice",
          "property": "b",
          "name": "ALICE"
        }
      }
    ],
    "edges": [
      {
        "kind": "Knows",
        "start": {
          "value": "123",
          "match_by": "id"
        },
        "end": {
          "value": "234",
          "match_by": "id"
        }
      }
    ]
  }
}
```

```
{
  "graph": {
    "nodes": [
      {
        "id": "123",
        "kinds": [
          "Person",
          "Base"
        ],
        "properties": {
          "displayname": "bob",
          "property": "a",
          "name": "BOB"
        }
      },
      {
        "id": "234",
        "kinds": [
          "Person",
          "Base"
        ],
        "properties": {
          "displayname": "alice",
          "property": "b",
          "name": "ALICE"
        }
```

```
18          "kinds": [
19            "Person",
20            "Base"
21          ],
22          "properties": {
23            "displayname": "alice",
24            "property": "b",
25            "name": "ALICE"
26          }
27        }
28      ],
29      "edges": [
30        {
31          "kind": "Knows",
32          "start": {
33            "value": "123",
34            "match_by": "id"
35          },
36          "end": {
37            "value": "234",
38            "match_by": "id"
39          }
40        }
41      ]
42    }
43  }
```

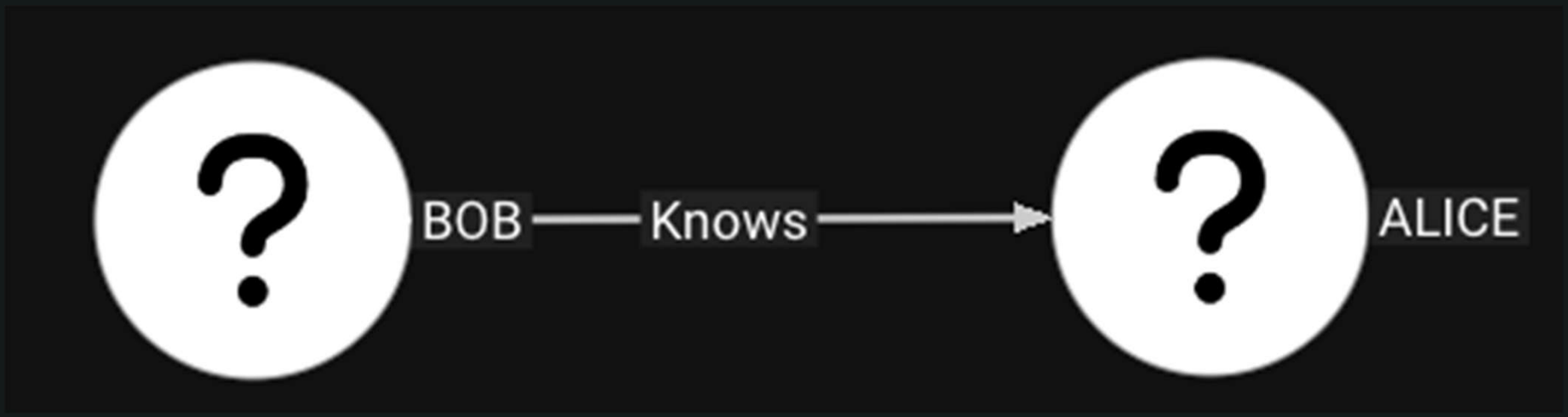
U/L data


Data Collection
File Ingest
Data Quality
Database Management
Users
Manage Users
Authentication
SSO Configuratio
Configuration
BloodHound Con
Early Access Fea
File Ingest
Upload data from SharpHound or AzureHound offline collectors. Check out our Getting Started documentation for more information.
Upload File(s)
Status
Status Messag
Complete
Complet
Complete
Complet
1–2 of 2
Click here or drag and drop to upload files
Cancel
Upload

Use Cypher to look at the data

Cypher is the Graph Database language.

```
match p=()-[:Knows]->()
return p

match p=(:Person)-[]->()
return p
```


BOB
Knows
ALICE

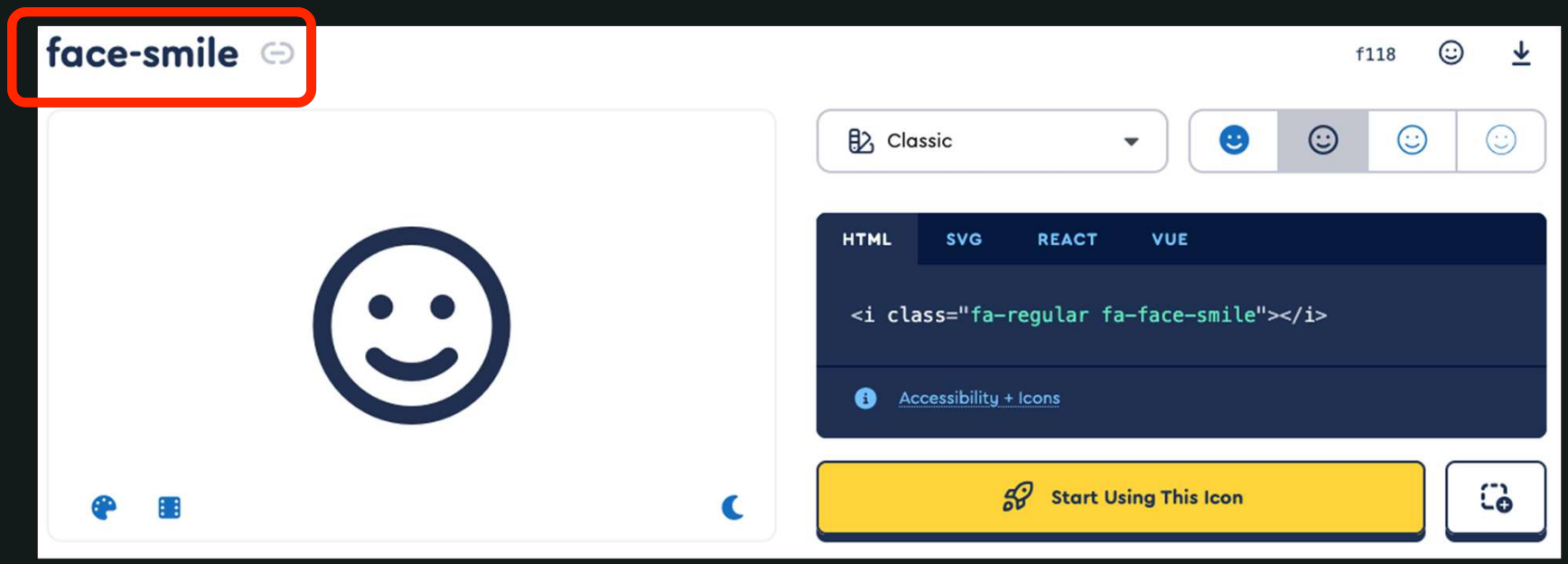
Pathfinding

Show paths between 2 nodes

Add custom icons

Because nobody likes (?)

Fontawesome.com

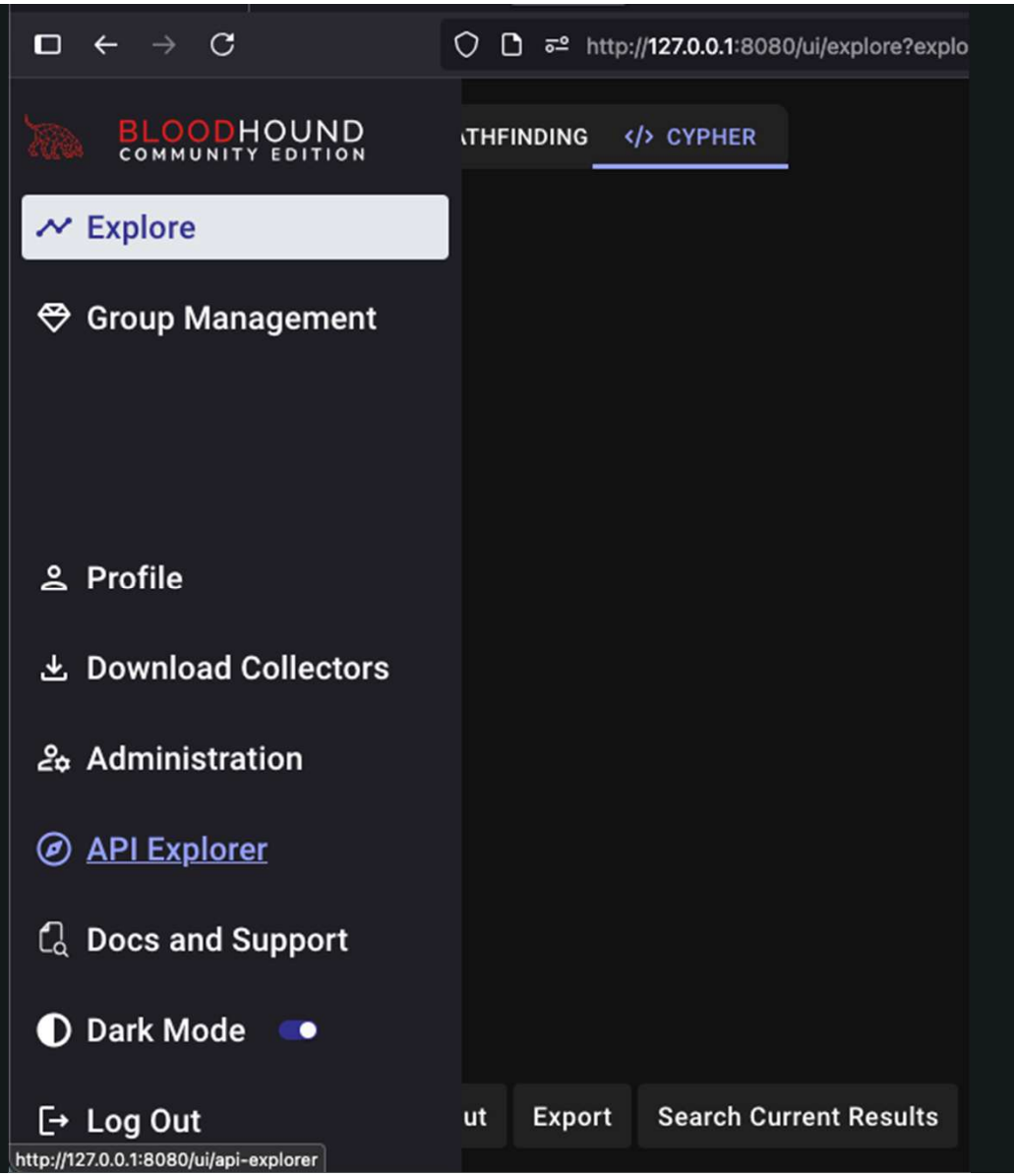


```python
import requests
import json
import urllib3

urllib3.disable_warnings(urllib3.exceptions.InsecureRequestWarning)

url = "http://127.0.0.1:8585/api/v2/custom-nodes"

headers = {
    "Authorization": "Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.e
    "Content-Type": "application/json"
}

def define_icon(icon_type, icon_name, icon_color):
    payload = {
        "custom_types": {
            icon_type: {
                "icon": {
                    "type": "font-awesome",
                    "name": icon_name,
                    "color": icon_color
                }
            }
        }
    }

    response = requests.post(
        url,
        headers=headers,
        json=payload,
        verify=False  # Disables SSL verification
    )

    print(f"🔹 Sent icon for: {icon_type}")
    print("Status Code:", response.status_code)
    print("Response Body:", response.text)
    print("---")

# Call function for each icon type you want to send
define_icon("Person", "face-smile", "#03CEFC")
```

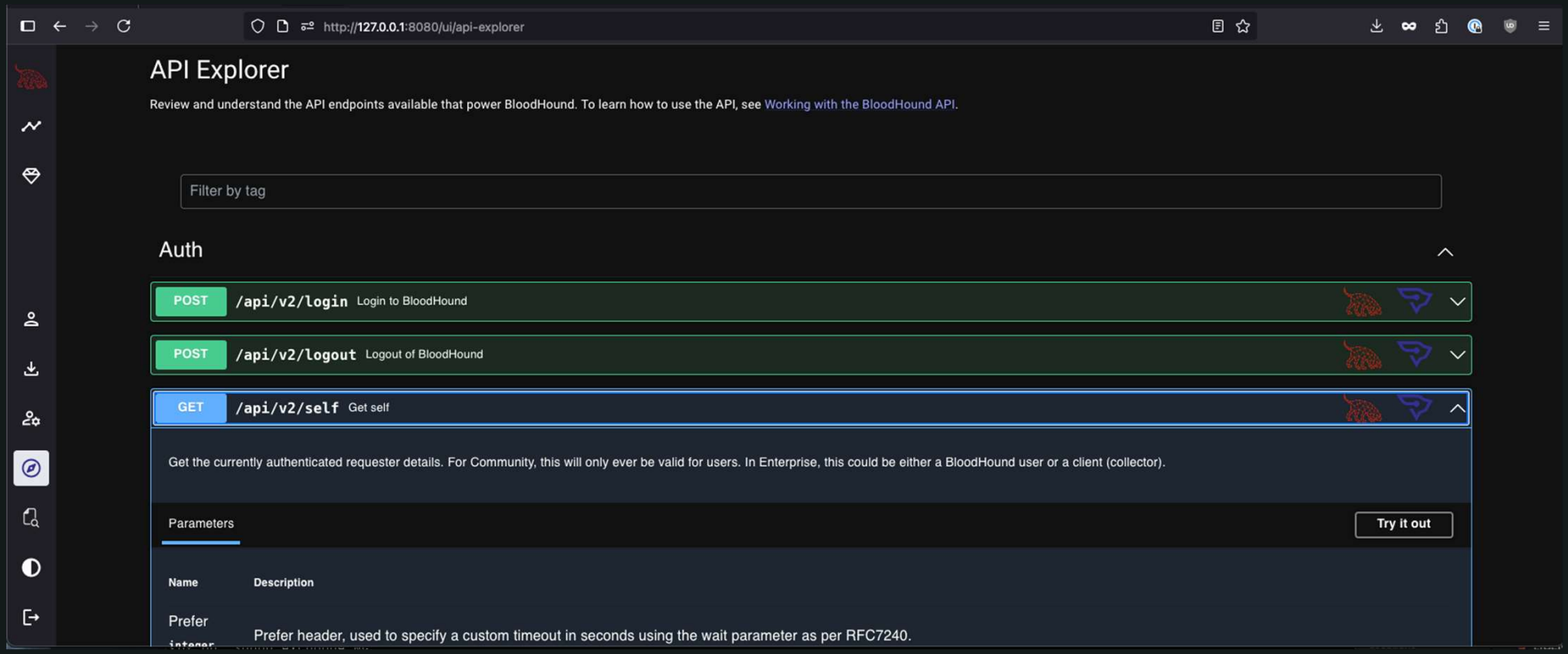
```python

url = "http://127.0.0.1:8585/api/v2/custom-nodes"

headers = {
    "Authorization": "Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.e
    "Content-Type": "application/json"
}

def define_icon(icon_type, icon_name, icon_color):
    payload = {
        "custom_types": {
            icon_type: {
                "icon": {
                    "type": "font-awesome",
                    "name": icon_name,
                    "color": icon_color
                }
            }
        }
    }

    response = requests.post(
        url
```

```python
                    "type": "font-awesome",
                    "name": icon_name,
                    "color": icon_color
                }
            }
        }
    }

    response = requests.post(
        url,
        headers=headers,
        json=payload,
        verify=False  # Disables SSL verification
    )

    print(f"🔷 Sent icon for: {icon_type}")
    print("Status Code:", response.status_code)
    print("Response Body:", response.text)
    print("---")

# Call function for each icon type you want to send
define_icon("Person", "face-smile", "#03CEFC")
```

Bearer Token



GET (Self)


http://127.0.0.1:8080/ui/api-explorer
API Explorer
Review and understand the API endpoints available that power BloodHound. To learn how to use the API, see Working with the BloodHound API.
Filter by tag
Auth
POST /api/v2/login Login to BloodHound
POST /api/v2/logout Logout of BloodHound
GET /api/v2/self Get self
Get the currently authenticated requester details. For Community, this will only ever be valid for users. In Enterprise, this could be either a BloodHound user or a client (collector).
Parameters
Try it out
Name
Description
Prefer
Prefer header, used to specify a custom timeout in seconds using the wait parameter as per RFC7240.

GET (Self)

http://127.0.0.1:8080/ui/api-explorer

GET /api/v2/self Get self

Get the currently authenticated requester details. For Community, this will only ever be valid for users. In Enterprise, this could be either a BloodHound user or a client (collector).

Parameters

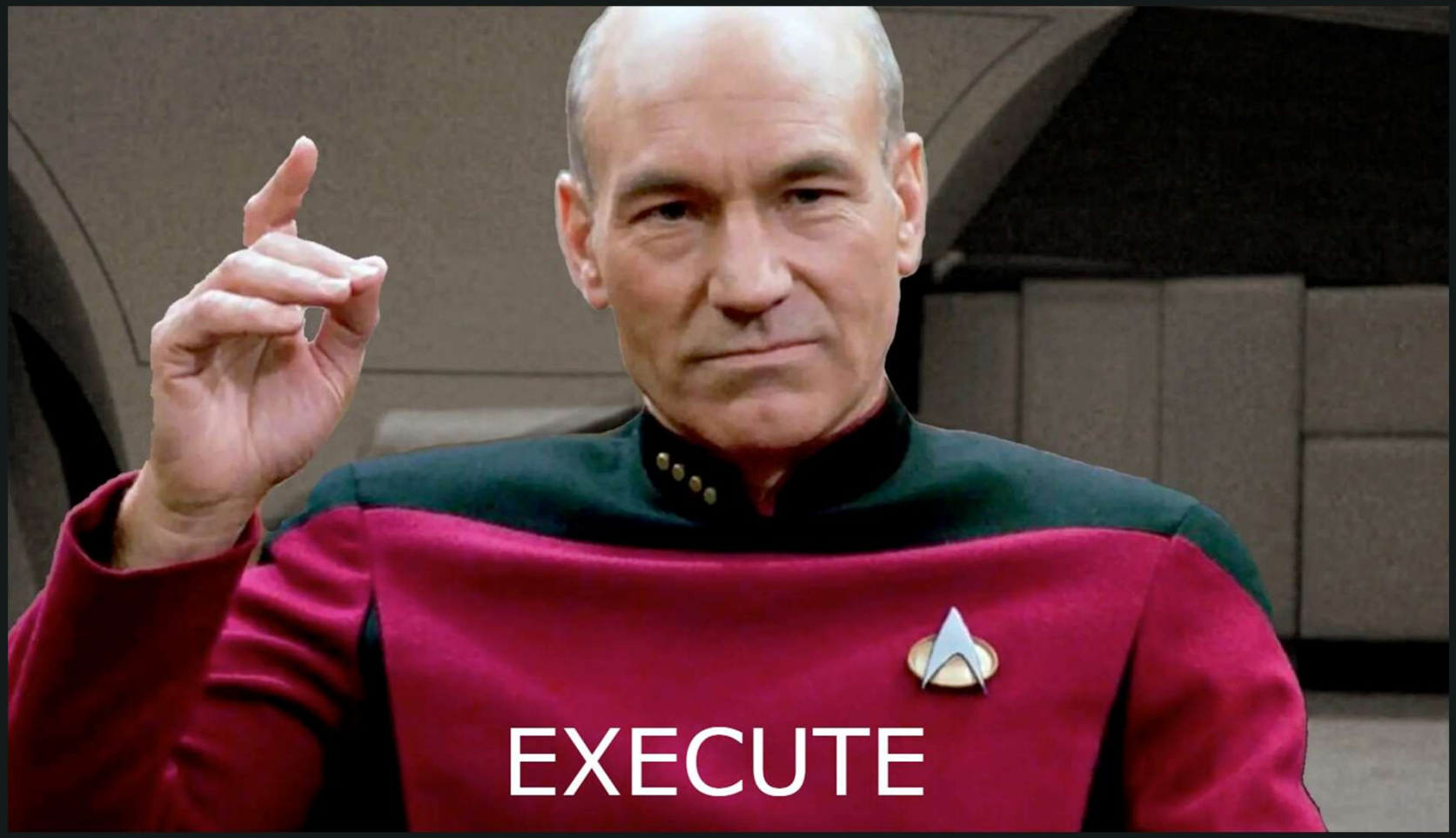
Cancel

Name	Description
Prefer integer (header)	Prefer header, used to specify a custom timeout in seconds using the wait parameter as per RFC7240. 0

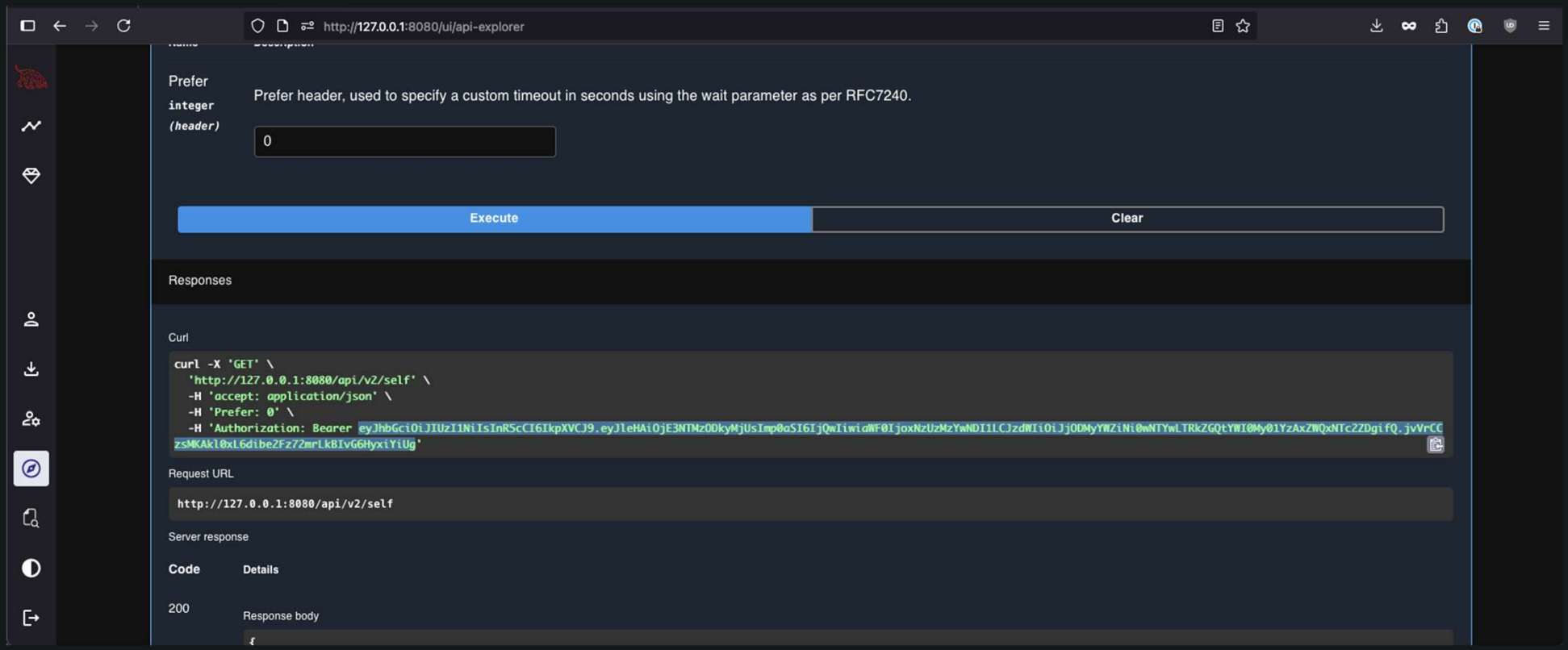
Execute

Responses

Code	Description	Links
200	OK Media type	No links


EXECUTE

Copy


http://127.0.0.1:8080/ui/api-explorer
Prefer
integer
(header)
Prefer header, used to specify a custom timeout in seconds using the wait parameter as per RFC7240.
0
Execute
Clear
Responses
Curl
curl -X 'GET' \
'http://127.0.0.1:8080/api/v2/self' \
-H 'accept: application/json' \
-H 'Prefer: 0' \
-H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJleHAiOjE3NTMzODkyMjUsImp0aSI6IjQwIiwiaWF0IjoxNzUzMzYwNDI1LCJzdWIiOiJjODMyYWZiNi0wNTYwLTRkZGQtYWI0My01YzAxZWQxNTczZDgifQ.jvVrCCzsMKAkl0xL6dibeZFz7ZmrLkBIvG6HyxiYiUg'
Request URL
http://127.0.0.1:8080/api/v2/self
Server response
Code
Details
200
Response body

API Explorer

Review and understand the API endpoints available that power BloodHound. To learn how to use the API, see Working with the BloodHound API.

cust

Custom Node Management

GET /api/v2/custom-nodes Get custom nodes

POST /api/v2/custom-nodes Create custom nodes

This endpoint registers new node kinds along with optional display metadata such as icons and colors. The provided configuration controls how nodes of the specified kinds are interpreted and rendered in the UI. Only free, solid-style Font Awesome icons are supported. Hex color codes must follow the format #RGB or #RRGGBB. If no valid icon or color is provided, the system will fall back to a question mark icon and a white background color. Batch input is supported; you may submit multiple configuration objects in a single request.

Parameters

Try it out

Name	Description
Prefer integer (header)	Prefer header, used to specify a custom timeout in seconds using the wait parameter as per RFC7240. *Default value* : 0 0

Request body required

application/json

POST `/api/v2/custom-nodes` Create custom nodes

This endpoint registers new node kinds along with optional display metadata such as icons and colors. The provided configuration controls how nodes of the specified kinds are interpreted and rendered in the UI. Only free, solid-style Font Awesome icons are supported. Hex color codes must follow the format #RGB or #RRGGBB. If no valid icon or color is provided, the system will fall back to a question mark icon and a white background color. Batch input is supported; you may submit multiple configuration objects in a single request.

Parameters

Cancel

Name	Description
Prefer *integer* *(header)*	Prefer header, used to specify a custom timeout in seconds using the wait parameter as per RFC7240. 0

Request body required

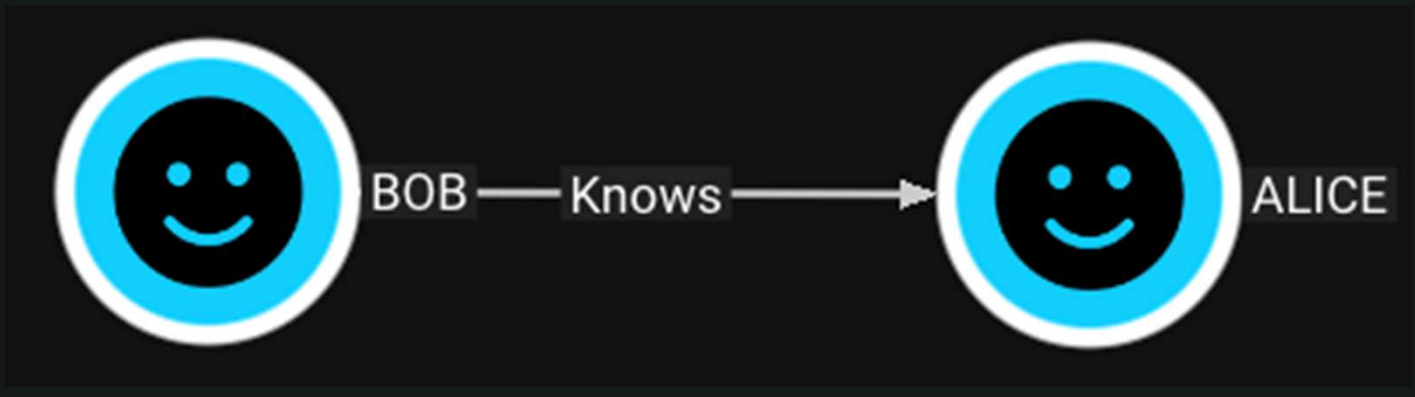
application/json

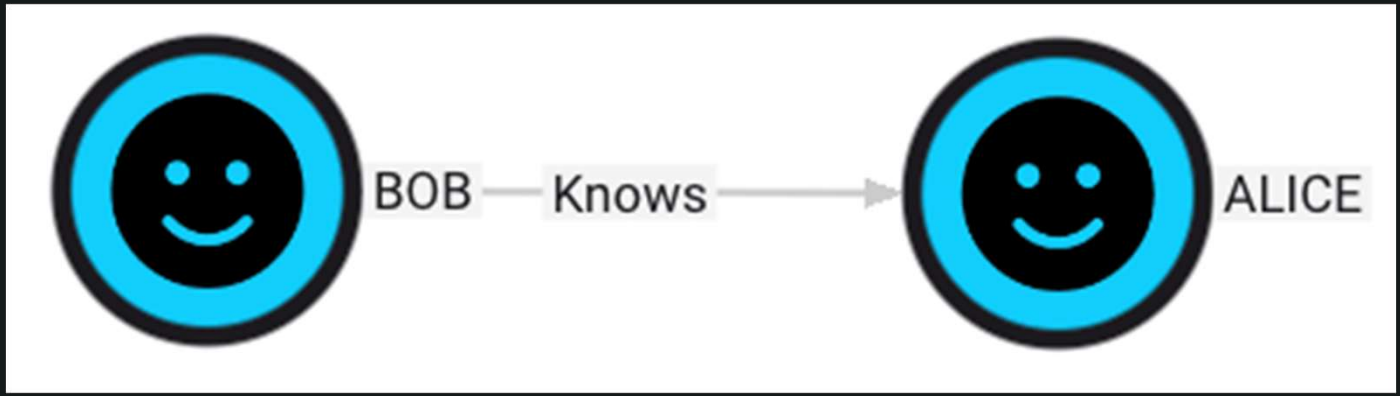
```
{
  "custom_types": {
    "KindA": {
      "icon": {
        "type": "font-awesome",
        "name": "house",
        "color": "#FFFFFF"
      }
    },
    "KindB": {
      "icon": {
        "type": "font-awesome",
        "name": "coffee",
        "color": "#000000"
      }
    }
  }
}
```

Execute

Responses

```
{
  "custom_types": {
    "Person": {
      "icon": {
        "type": "font-awesome",
        "name": "face-smile",
        "color": "#03CEFC"
      }
    }
  }
}
```


BOB
Knows
ALICE


BOB
Knows
ALICE

```json
{
  "schema": {
    "name": "OG",
    "display_name": "OG",
    "version": "v0.0.1",
    "namespace": "OG"
  },
  "node_kinds": [
    {
      "name": "OG_Person",
      "display_name": "Person",
      "description": "",
      "is_display_kind": true,
      "icon": "face-smile",
      "color": "#03CEFC"
    }
  ],
  "relationship_kinds": [
```

```
8   "node_kinds": [
9     {
10      "name": "OG_Person",
11      "display_name": "Person",
12      "description": "",
13      "is_display_kind": true,
14      "icon": "face-smile",
15      "color": "#03CEFC"
16    }
17  ],
18  "relationship_kinds": [
19    {
20      "name": "OG_Knows",
21      "description": "",
22      "is_traversable": true
23    }
24  ]
25 }
```

Helper Tools

BH Operator

bhopengraph/DataHound

gopengraph

OpenHound

LAB 1

Create a JSON
Query the Graph
Define Schema

https://github.com/Mayyhem/OpenGraphWorkshop

LAB 1

ogworkshop.bhe.io

Username: user@og.workshop
Password: OpenGraphWorksh0p!

Model Your Attack Paths

The Most
Important Part

In Active Directory

User

Domain Admins (Group)

Arrows.app

Let you iterate fast

arrows.app/#/local/id=Le4kMxxgPjbqIpIQj6pT

Untitled graph

Saved to Web Browser storage

Download / Export

Save to Google Drive

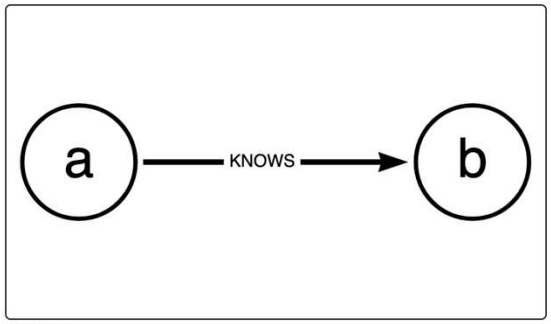
Graph:

nodes: 1

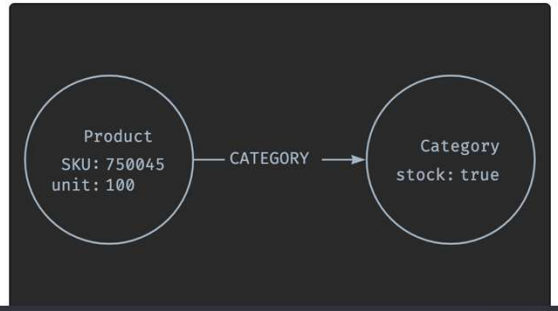
Add Node

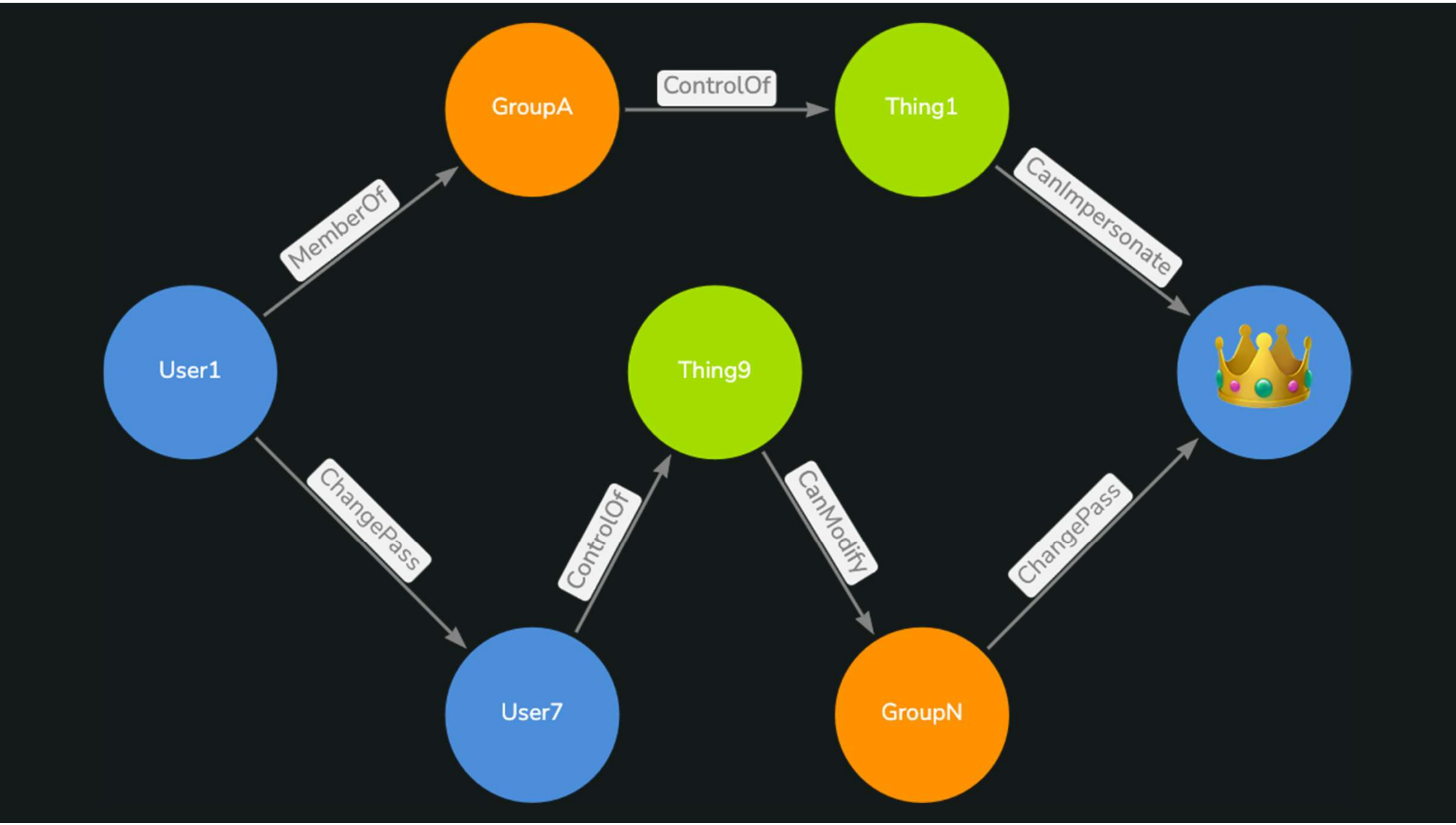
STYLE

Theme | Customize



Chunky

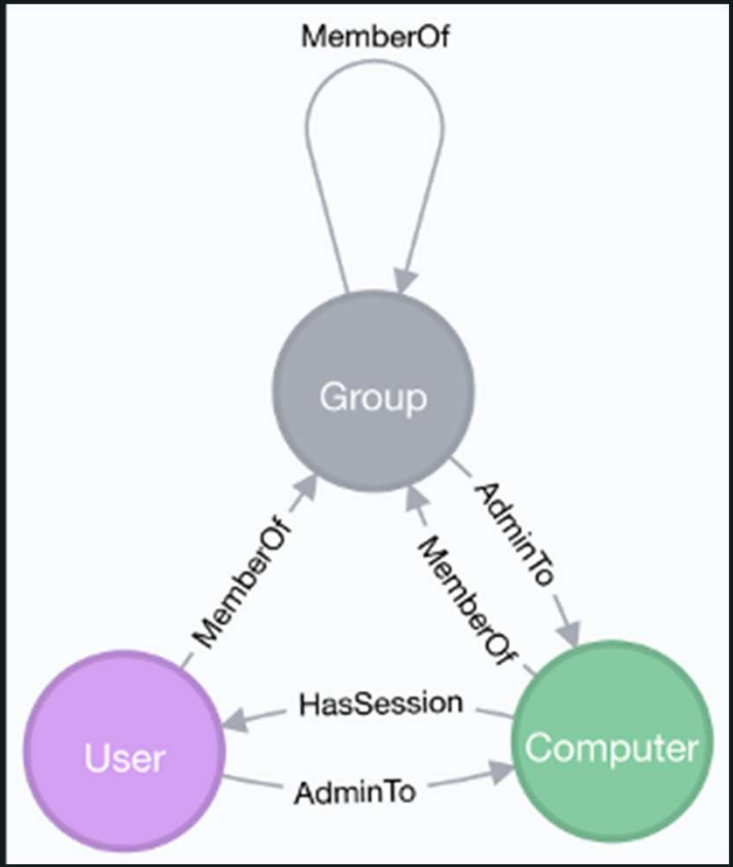



GroupA
ControlOf
Thing1
MemberOf
CanImpersonate
User1
Thing9
ChangePass
ControlOf
CanModify
ChangePass
User7
GroupN

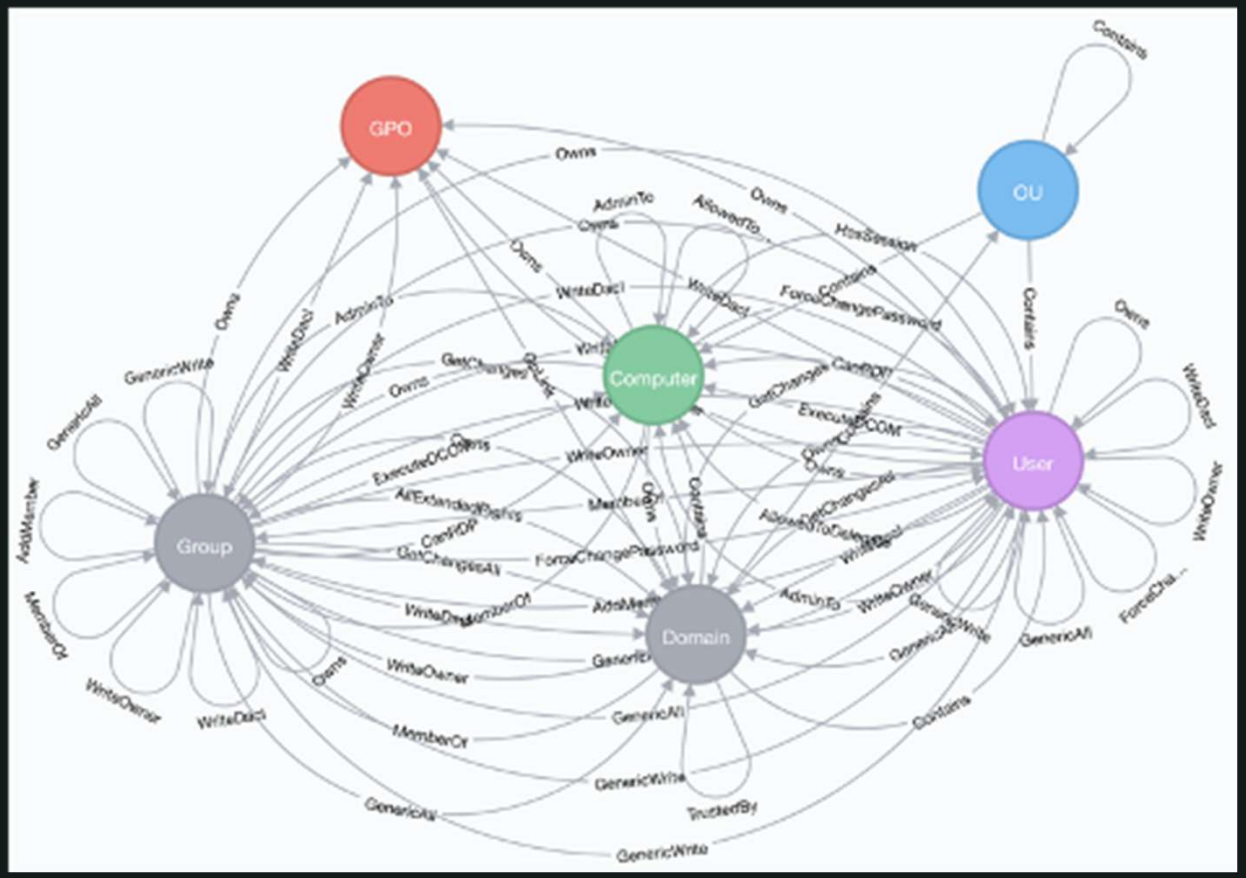
Perfect is the enemy of good

© Voltaire (1770)

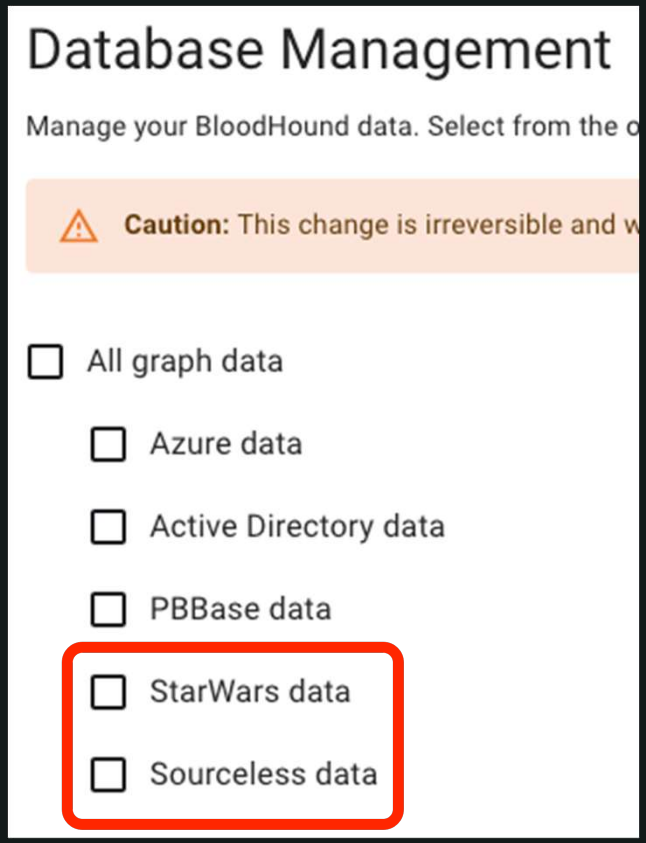
BloodHound 1.0



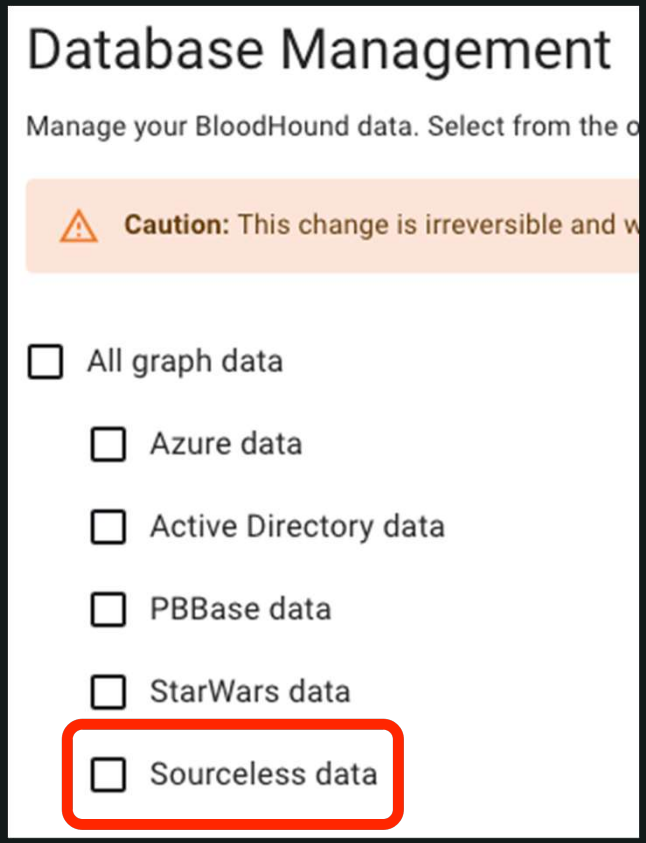
BloodHound 2.0



Clear OpenGraph Data



Clear OpenGraph Data



Clear OpenGraph Data

```
{
  "metadata": {
      "source_kind": "StarWars"
  },
  "graph": {
    "nodes": [
```

Graph Anything

Access vs Attack Graph

ork

Google Maps

Entry and Exit Points

Tech Support

Force Change Password

Domain Admins

Edge Traversability

Not all Edges are created equal

Traversable Edge

A has complete control of B

A can follow all attack paths from B

Non-Traversable Edge

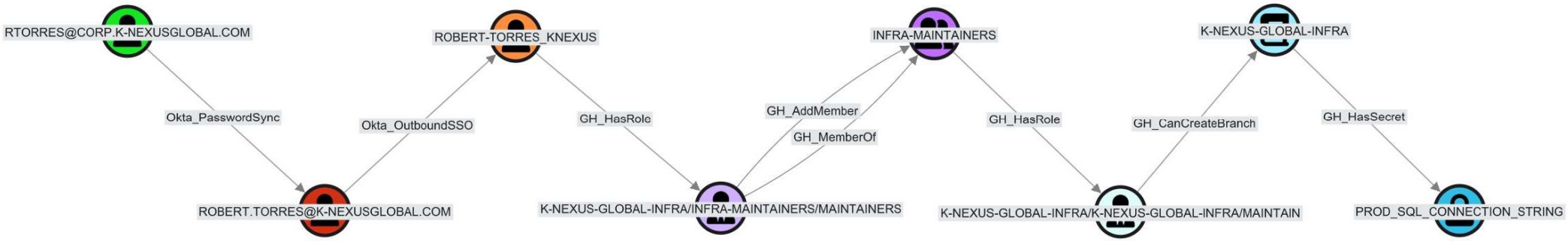
A has some control of B

A can not follow all attack paths from B

Hybrid Edges

The True Strength of OG

Hybrid Attack Path



More on Graph Theory

bloodhound.specterops.io/opengraph/graph-theory

All the Hounds

bloodhound.specterops.io/opengraph/library

OpenGraph Library

- 1Password
- Ansible
- GitHub
- JamfHound
- MSSQL
- Snowflake
- vCenter

- 1Password
- Ansible
- Active Directory (AD)
- Amazon Web Service (AWS)
- Atlassian
- Cisco Duo Security
- Credentials
- CyberArk
- Entra ID
- FreeIPA
- GitHub
- Google Cloud Platform (GCP)
- Jamf
- MSSQL

- GitHub
- Google Cloud Platform (GCP)
- Jamf
- MSSQL
- Network
- Okta
- Oracle Cloud Infrastructure
- Ping
- runZero
- Salesforce
- Snowflake
- System Center
- vCenter
- Windows

LAB 2

Collecting nodes and edges

Expanding the graph

Connecting two graphs (hybrid edges)

Contribute

Share your Hounds!

Mandatory

A Collector (Hound)

Documentation

- Minimum Requirements
- How-to
  - Install
  - Run

Set your project appart

- Node & Edge Information
- Cypher Queries
- Icon Pack
- Arrows.app Diagram

What’s in it for me?

OCIP

Write a Blog

Give a Talk

Leaving on a Jet Plane
Financially Blonde

More on Incentive Program

bloodhound.specterops.io/opengraph/ocip

ProTip

Enable Graph Mutation via Cypher

```
67    bloodhound:
68      labels:
69        name: bhce_bloodhound
70      image: docker.io/specterops/bloodhound:${BLOODHOUND_TAG:-edge}
71      environment:
72        - bhe_disable_cypher_complexity_limit=${bhe_disable_cypher_complexity
73        - bhe_enable_cypher_mutations=${bhe_enable_cypher_mutations:-true}
74        - bhe_graph_query_memory_limit=${bhe_graph_query_memory_limit:-2}
75        - bhe_database_connection=user=${POSTGRES_USER:-bloodhound} password=
76        - bhe_neo4j_connection=neo4j://${NEO4J_USER:-neo4j}:${NEO4J_SECRET:-b
77        - bhe_recreate_default_admin=${bhe_recreate_default_admin:-false}
78        - bhe_enable_text_logger=${bhe_enable_text_logger:-true}
79        ### Add additional environment variables you wish to use here.
```

Delete Edges

```
match p=()-[r:Knows]->()
delete r
```

Delete Nodes

```
match (m:Person)
where m.name = "BOB"
delete m
```

Open Hours

Monday & Tuesday
13h15
29th Floor
Southwest Corner

Workshop: Build Your Own OpenGraph Collector

@_MAYYHEM

linkedin.com/in/X24

@ScoubiMtl

linkedin.com/in/mathieusaulnier